JNTUH UNIVERSITY COLLEGE OF ENGINEERING JAGTIAL

DEPARTMENT OF ELECTRONIC AND COMMUNICATION ENGINEERING

Title of the project :

ACCIDENT PREVENTION SYSTEM AT BLIND ROAD TURN

ABSTRACT

This project includes the design and development of an IoT-based accident prevention system at blind road turns using an Arduino microcontroller. Blind road turns are places where drivers cannot clearly see vehicles coming from the opposite direction because of curves, buildings, trees, or hills. Due to this lack of visibility, drivers may suddenly face another vehicle when they reach the turn, which can lead to serious road accidents. Many accidents happen especially on village roads, hill roads, and narrow streets where drivers cannot predict the movement of vehicles from the other side. Therefore, there is a need for a simple system that can warn drivers before they reach a blind turn.

In this project, a smart warning system is designed using basic embedded systems and IoT technology to reduce accidents. The system mainly uses Arduino as the main controller, along with sensors and alert devices.

The working of the project is simple and easy to understand. First, vehicle detection sensors are placed on both sides of the blind road turn. These sensors continuously monitor whether a vehicle is approaching the turn. When a vehicle comes near the turn on one side, the sensor detects the vehicle and sends a signal to the Arduino microcontroller. The Arduino processes this signal and immediately activates a warning system on the opposite side of the road. The IoT module connected to the Arduino can transmit information such as vehicle detection alerts to a mobile application or cloud platform.

Overall, this project aims to reduce road accidents at blind road turns by providing a smart vehicle detection and warning system, making roads safer for drivers and pedestrians. The system demonstrates how embedded systems and IoT technology can be used to solve real-world problems in transportation safety

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